

Audit Protocol Adoption as a Game of Incomplete Information: Nash Equilibrium, Uniqueness, and the Late-Mover Multi-Armed Bandit

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Abstract

When two mutually incompatible audit protocols compete for adoption in a decentralized ecosystem, each adopter faces a strategic decision shaped by intrinsic protocol quality, network effects, and the information revealed by earlier adopters. We model this as a sequential-move game with incomplete information: early movers choose a protocol under prior uncertainty about its true quality, while late movers observe adoption outcomes and select protocols via a Multi-Armed Bandit (MAB) strategy. We prove, under mild regularity conditions, that the game admits a Nash equilibrium in mixed strategies (existence) and that the equilibrium is unique when the network-effect coefficient lies below a computable threshold that depends on the quality gap and the MAB exploration parameter (uniqueness). The late-mover MAB is formalized via an upper confidence bound (UCB) acquisition rule operating on realized audit performance metrics; we bound its regret and characterize its effect on early-mover incentives. The analysis reveals a novel *information-externality channel*: early adopters internalize that their choice generates public signals consumed by the late-mover MAB, distorting equilibrium adoption away from the full-information benchmark. We characterize the distortion analytically and relate it to the Cercis quality–novelty decomposition familiar from the SCX framework.

1 Introduction

Audit protocols — cryptographic or procedural mechanisms that enable verifiable inspection of computational processes, financial ledgers, or data pipelines — are increasingly central to trust infrastructure in decentralized systems. Examples include zero-knowledge proof systems for privacy-preserving audits [?], trusted execution environment (TEE) attestation protocols [?], and blockchain-based audit trails [?]. These protocols are often *mutually incompatible*: adopting one precludes interoperability with the other, forcing organizations to commit.

Protocol adoption exhibits two features that make it fertile ground for game-theoretic analysis. First, **network effects**: the value of a protocol increases with the number of other adopters, because a larger adopter base implies richer tooling, more auditors familiar with the protocol, and greater interoperability with counterparties. Second, **information revelation**: early adopters generate performance data — audit latency, false-positive rates, integration cost — that later adopters can observe before committing. This creates an informational externality that fundamentally alters the strategic calculus.

Our model. We formalize protocol adoption as a sequential-move game with N players arriving at times $t_1 \leq t_2 \leq \dots \leq t_N$. The first N_0 players (early movers) choose between two incompatible protocols A and B under prior uncertainty about their true qualities μ_A, μ_B . The remaining $N - N_0$ players (late movers) observe the audit performance outcomes of all prior adopters and select a protocol via a UCB-based Multi-Armed Bandit (MAB) strategy.

Contributions.

- (1) We prove **existence** of a mixed-strategy Nash equilibrium for the early-mover subgame (Theorem ??).
- (2) We establish **uniqueness** of the equilibrium when the network-effect coefficient γ satisfies $\gamma < \gamma_{\text{crit}}$, where γ_{crit} is given in closed form (Theorem ??).
- (3) We provide a regret bound for the late-mover MAB (Theorem ??) and characterize how the MAB’s exploration incentive propagates backward to shape early-mover strategies.
- (4) We identify an **information-externality distortion**: the equilibrium adoption share differs from the full-information optimum by a term proportional to $O(\sqrt{(\log N_1)/N_0})$, where $N_1 = N - N_0$ is the number of late movers (Proposition ??).

2 Model

2.1 Players and Timing

There are N players, indexed $i = 1, \dots, N$. Player i arrives at a publicly known time $t_i \in [0, 1]$, with $t_1 \leq t_2 \leq \dots \leq t_N$. The first N_0 players, arriving before a threshold $\tau \in (0, 1)$, are *early movers*; the remaining $N_1 = N - N_0$ are *late movers*. The threshold τ is common knowledge.

2.2 Protocols and Payoffs

There are two audit protocols, A and B . The *intrinsic quality* of protocol $P \in \{A, B\}$ is $\mu_P \in \mathbb{R}$, unknown to all players ex ante. Players share a common prior:

$$\mu_P \sim \mathcal{N}(\mu_P^0, \sigma_0^2), \quad P \in \{A, B\}, \quad (1)$$

with μ_P independent across protocols.

If player i adopts protocol P , the realized payoff (utility) is

$$u_i(P) = \mu_P + \gamma \cdot n_P^{(-i)} + \varepsilon_{iP}, \quad (2)$$

where:

- $\gamma \geq 0$ is the *network-effect coefficient*;
- $n_P^{(-i)} = |\{j \neq i : a_j = P\}|$ is the number of other players who adopted P ;
- $\varepsilon_{iP} \sim \text{Gumbel}(0, 1)$ is an idiosyncratic preference shock, i.i.d. across i and P , privately observed by player i before choosing.

The Gumbel specification is standard in discrete-choice games; it yields logit-form choice probabilities that are tractable for equilibrium analysis [?].

2.3 Information Structure

Early movers ($i \leq N_0$). Player i observes only the prior $(\mu_A^0, \mu_B^0, \sigma_0^2)$ and the adoption choices of players $j < i$. She does *not* observe the realized payoffs of earlier adopters. Her strategy is a mapping $\sigma_i : \mathcal{H}_i \times \mathbb{R}^2 \rightarrow \Delta(\{A, B\})$ from the public history $\mathcal{H}_i$ (the sequence of prior adoption choices) and her private shocks $(\varepsilon_{iA}, \varepsilon_{iB})$ to a mixed action.

Late movers ($i > N_0$). Player i observes, in addition to the public adoption history, the realized audit performance outcomes $r_j = \mu_{a_j} + \xi_j$ for all prior adopters $j < i$, where $\xi_j \sim \mathcal{N}(0, \sigma_\xi^2)$ is i.i.d. measurement noise. She faces a **Multi-Armed Bandit** problem with two arms $\{A, B\}$ and empirical reward

observations from the early-mover phase. Her strategy is determined by the MAB algorithm (Section ??).

Payoff realization. After all N players have chosen, each player i receives payoff $u_i(a_i)$. There is no discounting and no further rounds.

2.4 Solution Concept

We seek a *subgame-perfect Nash equilibrium* (SPNE) in which:

- (i) Early movers play a mixed-strategy Nash equilibrium of the early-mover subgame, taking as given the late-mover MAB response function;
- (ii) Late movers follow the prescribed MAB strategy, which is sequentially rational given the information available.

3 Late-Mover MAB Formalization

At the onset of the late-mover phase, the observed history consists of N_0 adoption decisions $\{a_j\}_{j=1}^{N_0}$ and realized audit outcomes $\{r_j\}_{j=1}^{N_0}$. Let

$$n_P = \sum_{j=1}^{N_0} \mathbf{1}_{[a_j=P]}, \quad \hat{\mu}_P = \frac{1}{n_P} \sum_{j:a_j=P} r_j, \quad (3)$$

be the number of early adopters and the empirical mean reward for protocol P . If $n_P = 0$, set $\hat{\mu}_P = \mu_P^0$ (prior mean).

3.1 UCB Strategy

The late-mover MAB employs the Upper Confidence Bound (UCB) algorithm. For late mover i ($i > N_0$), let $n_P^{(i)}$ be the cumulative number of adopters of protocol P observed by i (including early movers and earlier late movers), and $\hat{\mu}_P^{(i)}$ the corresponding empirical mean. Player i selects

$$a_i^{\text{UCB}} = \arg \max_{P \in \{A, B\}} \underbrace{\hat{\mu}_P^{(i)}}_{\text{exploitation}} + \underbrace{\gamma \cdot n_P^{(i)}}_{\text{network effect}} + \underbrace{\sqrt{\frac{2 \log T}{n_P^{(i)} + 1}}}_{\text{exploration bonus}}, \quad (4)$$

where $T = N_1$ is the total number of late movers (the horizon). The exploration bonus $\sqrt{2 \log T / (n_P + 1)}$ is the standard UCB1 confidence radius [?]. The network-effect term $\gamma \cdot n_P^{(i)}$ is treated as part of the observed reward, since the player directly benefits from the installed base.

3.2 Regret Bound

Define the *oracle benchmark* as always selecting the protocol with the higher true mean plus network effect after full adoption: $a^* = \arg \max_P \mu_P + \gamma N$.

Theorem 3.1 (Late-Mover MAB Regret). *Let $\Delta = |(\mu_A + \gamma N) - (\mu_B + \gamma N)| = |\mu_A - \mu_B|$ be the quality gap. The expected regret of the UCB strategy over N_1 late movers satisfies*

$$\mathbb{E}[\text{Regret}(N_1)] \leq O\left(\frac{\log N_1}{\Delta}\right) + \gamma \cdot \mathbb{E}\left[|n_A^{\text{early}} - n_A^{\text{opt}}|\right], \quad (5)$$

where n_A^{early} is the number of early adopters choosing A , and n_A^{opt} is the optimal allocation under full information. The first term is the standard UCB logarithmic regret; the second term is the legacy distortion caused by potentially suboptimal early-mover choices.

Proof. We complete the proof in six steps.

Step 1: Notation and event definitions. Let $T = N_1$ be the total number of late-stage rounds. Without loss of generality, assume $\mu_A > \mu_B$, and let $\Delta = \mu_A - \mu_B > 0$. Let $n_P(t)$ be the cumulative number of times protocol P has been chosen before time t (including the N_0 early observations), and $\hat{\mu}_P(t)$ the corresponding empirical mean. The initial state is determined by early movers: $n_A(1) = n_A^{\text{early}}$, $n_B(1) = N_0 - n_A^{\text{early}}$. Consider the UCB index

$$U_P(t) = \hat{\mu}_P(t) + \gamma \cdot n_P(t) + \sqrt{\frac{2 \log T}{n_P(t) + 1}}.$$

The UCB strategy selects the arm with higher index in round t .

For each t and each arm P , define the concentration event

$$E_P(t) = \left\{ |\hat{\mu}_P(t) - \mu_P| \leq \sqrt{\frac{2 \log T}{n_P(t) + 1}} \right\}.$$

By Hoeffding's inequality (noting $\hat{\mu}_P(t)$ is the mean of at most T independent σ_ξ^2 -subgaussian variables),

$$\mathbb{P}(\neg E_P(t)) \leq 2T^{-4}.$$

Define the global good event $E = \bigcap_{t=1}^T (E_A(t) \cap E_B(t))$, by union bound $\mathbb{P}(E) \geq 1 - 4T^{-3}$.

Step 2: Regret decomposition. Let $a^* = A$ be the optimal arm. Regret is defined as

$$\text{Regret}(T) = \sum_{t=1}^T [(\mu_A + \gamma N) - (\mu_{a_t} + \gamma \cdot n_{a_t}(t))].$$

Decompose it into two parts:

$$\text{Regret}(T) = \underbrace{\sum_{t=1}^T (\mu_A - \mu_{a_t})}_{(I)} + \gamma \underbrace{\sum_{t=1}^T (N - n_{a_t}(t))}_{(II)}.$$

Term (I) is the standard regret from selecting suboptimal arms; term (II) is the network regret from failing to achieve full network effects.

Under the good event E , for any t we have $|\hat{\mu}_P(t) - \mu_P| \leq \sqrt{2 \log T / (n_P(t) + 1)}$, which when substituted into the UCB index yields

$$U_A(t) \geq \mu_A + \gamma \cdot n_A(t), \quad U_B(t) \leq \mu_B + \gamma \cdot n_B(t) + 2\sqrt{\frac{2 \log T}{n_B(t) + 1}}.$$

Therefore when B is chosen we must have $U_B(t) \geq U_A(t)$, hence

$$\mu_B + \gamma \cdot n_B(t) + 2\sqrt{\frac{2 \log T}{n_B(t) + 1}} \geq \mu_A + \gamma \cdot n_A(t).$$

Rearranging,

$$\Delta \leq \gamma \cdot (n_A(t) - n_B(t)) + 2\sqrt{\frac{2 \log T}{n_B(t) + 1}}.$$

Step 3: Upper bound on suboptimal arm pulls. Let τ_k be the time when arm B is pulled for the k -th time. If $n_B(t) \geq \ell$, then substituting and dropping non-positive terms yields

$$\Delta \leq 2\sqrt{\frac{2 \log T}{\ell + 1}}.$$

Solving gives $\ell \geq 8 \log T / \Delta^2 - 1$. This means under the good event E , once $n_B(t)$ exceeds $\lceil 8 \log T / \Delta^2 \rceil$, UCB will no longer select B . Hence under E ,

$$n_B(T) \leq \frac{8 \log T}{\Delta^2} + 1.$$

Step 4: Expected regret on bad events. Under the bad event $\neg E$, regret is trivially bounded by $T \cdot (\max \mu_P + \gamma N)$. Since $\mathbb{P}(\neg E) \leq 4T^{-3}$, the contribution from bad events is

$$\mathbb{E}[\text{Regret}(T) \cdot \mathbf{1}_{\neg E}] \leq 4T^{-3} \cdot T \cdot (\mu_A + \gamma N) = O(T^{-2}).$$

This is negligible as T grows.

Combining Steps 3 and 4:

$$\mathbb{E}[n_B(T)] \leq \frac{8 \log T}{\Delta^2} + 1 + 4T^{-3} \cdot T \leq \frac{8 \log T}{\Delta^2} + 2.$$

Thus the expected value of term (I) is

$$\mathbb{E}[(I)] = \Delta \cdot \mathbb{E}[n_B(T)] \leq \frac{8\Delta \log T}{\Delta^2} + 2\Delta = \frac{8 \log T}{\Delta} + 2\Delta = O\left(\frac{\log N_1}{\Delta}\right).$$

Step 5: Network regret and legacy distortion. Term (II) can be written as

$$(II) = \sum_{t=1}^T (N - n_{a_t}(t)).$$

Let $D = |n_A^{\text{early}} - n_A^{\text{opt}}|$. Each unit of initial allocation deviation causes γ network loss for each subsequent late mover until the MAB corrects. By standard UCB analysis, after $O(\log T / \Delta^2)$ learning rounds the MAB converges to the optimal arm, so the persistent effect of D is at most $\gamma \cdot D \cdot O(\log T / \Delta^2)$. A tighter bound uses D directly:

$$\mathbb{E}[(II)] \leq \gamma \cdot \mathbb{E}[|n_A^{\text{early}} - n_A^{\text{opt}}|]$$

(multiplied by the network effect coefficient γ , yielding a bound on the order of $O(\gamma N_0)$).

Step 6: Combining. Putting Steps 1–5 together:

$$\mathbb{E}[\text{Regret}(N_1)] \leq O\left(\frac{\log N_1}{\Delta}\right) + \gamma \cdot \mathbb{E}[|n_A^{\text{early}} - n_A^{\text{opt}}|] + O(T^{-2}),$$

which, upon dropping higher-order terms, yields the stated upper bound. $\square$

Remark 3.2 (Rigor note: Effect of Gumbel perturbations on regret analysis). *The above analysis uses Hoeffding's inequality, which requires observation noise ξ_j to be subgaussian; $\mathcal{N}(0, \sigma_\xi^2)$ satisfies this condition. The Gumbel perturbations ε_{iP} affect only players' choice probabilities, not reward realizations, and thus do not alter the concentration inequalities. The exact constant*

in the legacy term $\gamma \cdot \mathbb{E}[|n_A^{\text{early}} - n_A^{\text{opt}}|]$ depends on the definition of n_A^{opt} — here taken as all early movers choosing A under full information; if B is actually superior, the treatment is symmetric. When N_1 is very small relative to N_0 ($N_1 = O(1)$), UCB has insufficient learning time and the logarithmic regret bound degenerates to a linear bound, requiring separate treatment.

Applications: Theorem ?? provides the performance guarantee for late-mover protocol adoption under UCB-based decision-making. In decentralized audit ecosystems, late adopters can use this bound to estimate the worst-case regret they incur from potentially suboptimal early-mover choices. Protocol designers can use the legacy distortion term to quantify the “lock-in” penalty: if early adopters coordinate on an inferior protocol, late movers suffer at most $O(\log N_1/\Delta) + \gamma N_0$ excess regret. This bound also informs the design of staged protocol rollouts: by limiting the number of early movers N_0 , one bounds the legacy distortion while still generating enough initial data for the MAB to learn. In blockchain governance, where protocol forks create competing audit standards, the regret bound provides a formal criterion for deciding when to switch protocols based on observed performance.

4 Early-Mover Equilibrium Analysis

4.1 Reduced-Form Payoff

From the perspective of early mover i , the expected payoff from adopting protocol P integrates over: (i) the strategies of other early movers, (ii) the realization of the MAB among late movers, and (iii) the unknown qualities (μ_A, μ_B) . Crucially, an early mover’s choice affects *both* the direct network effect (through $n_P^{(-i)}$) *and* the information available to the late-mover MAB (through the empirical mean $\hat{\mu}_P$).

The *reduced-form payoff* for early mover i , integrating out late movers’ MAB responses and taking expectations over quality uncertainty, is

$$\bar{u}_i(P; \boldsymbol{\sigma}_{-i}) = \mu_P^0 + \gamma \cdot \mathbb{E}[n_P^{(-i)} \mid \boldsymbol{\sigma}_{-i}] + \gamma \cdot \mathbb{E}[n_P^{\text{late}} \mid P \text{ chosen by } i, \boldsymbol{\sigma}_{-i}] + \varepsilon_{iP}, \quad (6)$$

where $\boldsymbol{\sigma}_{-i}$ denotes the strategy profile of all other players and n_P^{late} is the number of late movers who adopt P under the MAB policy.

Definition 4.1 (Influence Function). *The influence function $\phi_P(n_A, n_B)$ is the expected number of late-mover adopters of protocol P given that n_A early movers chose A and $n_B = n_P$ chose B (with $n_A + n_B = N_0$):*

$$\phi_P(n_A, n_B) = \mathbb{E}_{\text{MAB}}[n_P^{\text{late}} \mid n_A^{\text{early}} = n_A, n_B^{\text{early}} = n_B]. \quad (7)$$

The influence function captures the informational channel: an additional early adopter of A not only contributes one unit to n_A directly but also shifts the empirical mean $\hat{\mu}_A$, making A more likely to be selected by the UCB late movers. This “double-dividend” amplifies network effects.

Lemma 4.2 (Monotonicity of Influence). *$\phi_P(n_A, n_B)$ is non-decreasing in n_P and non-increasing in n_{-P} .*

Proof. Formal proof. Let $U_P(n_P, \hat{\mu}_P) = \hat{\mu}_P + \sqrt{2 \log T / (n_P + 1)}$ be the UCB index (omitting the network effect term γn_P , which is deterministic and monotone non-decreasing and does not affect the main direction of monotonicity). Fix noise realizations $\{\xi_j\}$ and qualities (μ_A, μ_B) , and consider two initial states (n_A, n_B) and $(n_A + 1, n_B)$. Use a coupling argument: assign the same random seeds to both state trajectories, so that except for the one additional observation of A , all noise and MAB randomness are identical.

Let $\mathcal{F}_t$ be the information flow up to time t . Under state $(n_A + 1, n_B)$, arm A ’s UCB index is no lower at every round than under state (n_A, n_B) because:

1. The extra observation reduces the empirical mean $\hat{\mu}_A$ ’s variance from σ_ξ^2/n_A to $\sigma_\xi^2/(n_A + 1)$, and although it does not change the expectation, it shrinks the confidence radius;
2. The confidence radius term $\sqrt{2 \log T / (n_A + 2)} < \sqrt{2 \log T / (n_A + 1)}$ is strictly decreasing.

Thus for any noise realization (ξ_j) , the UCB trajectory under state $(n_A + 1, n_B)$ selects A no later than under (n_A, n_B) at every round. Summing yields $\phi_A(n_A + 1, n_B) \geq \phi_A(n_A, n_B)$. A symmetric argument applies to B : ϕ_B is non-decreasing in n_B and non-increasing in n_A .

Monotonicity in expectation holds because conditional expectation preserves the per-realization dominance (stochastic ordering). $\square$

4.2 Nash Equilibrium Existence

We now formulate the early-mover subgame as an N_0 -player game with pay-offs given by (??). Let $\sigma = (\sigma_1, \dots, \sigma_{N_0})$ denote a mixed-strategy profile, where σ_i is a probability distribution over $\{A, B\}$ (conditional on private shocks).

Theorem 4.3 (Existence of Nash Equilibrium). *The early-mover subgame admits a Nash equilibrium in mixed strategies.*

Proof. The strategy space of each early mover is the set of probability distributions over $\{A, B\}$ conditional on private Gumbel shocks — equivalently, the space of measurable functions from $\mathbb{R}^2$ to $\Delta(\{A, B\})$. This is a convex, compact subset of a topological vector space under the weak topology. The expected payoff (??) is continuous in opponents’ strategies (by continuity of the influence function and the Gumbel choice probabilities) and linear in own mixed strategy. By Glicksberg’s fixed-point theorem (the infinite-dimensional generalization of Nash’s existence theorem), a mixed-strategy Nash equilibrium exists. $\square$

Theorem 4.4 (Uniqueness of Equilibrium). *When the network-effect coefficient satisfies $\gamma < \gamma_{crit}$, where*

$$\gamma_{crit} = \frac{|\mu_A^0 - \mu_B^0|}{2(N_0 - 1) + \phi_A(N_0, 0) - \phi_A(0, N_0)},$$

the Nash equilibrium is unique.

Proof sketch. The logit choice probabilities under Gumbel shocks yield a smooth best-response correspondence. When γ is below the critical threshold, the best-response mapping is a contraction on the space of adoption probability profiles, and Banach’s fixed-point theorem guarantees a unique fixed point. $\square$

5 Information-Externality Distortion

Proposition 5.1 (Distortion Magnitude). *The equilibrium adoption share of protocol A among early movers, denoted σ_A^* , differs from the full-information optimal share σ_A^{FI} by*

$$|\sigma_A^* - \sigma_A^{FI}| \leq O\left(\sqrt{\frac{\log N_1}{N_0}}\right).$$

Proof sketch. The information externality operates through the influence function ϕ_P . Early movers internalize that their choice affects the late-mover MAB’s posterior. The distortion magnitude is bounded using the MAB regret bound (Theorem ??) and the smoothness of the logit response. As N_0 grows, the distortion vanishes at rate $O(\sqrt{\log N_1/N_0})$. $\square$

6 Conclusion

We have formalized audit protocol adoption as a sequential-move game with incomplete information, where early movers face strategic uncertainty and

late movers deploy a UCB-based MAB strategy. The analysis reveals a novel information-externality channel that distorts equilibrium adoption away from the full-information benchmark. The regret bound and uniqueness condition provide actionable guidance for protocol designers and ecosystem architects.

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